7.3 Multiplying and Dividing Rational Expressions

Name _____ Period ____ Date _____

Algebra II 7.3 (30-minute plan) · Student sheet

I can…

- simplify rational expressions and state the excluded values.
- multiply rational expressions.
- divide rational expressions.

Vocabulary · fill in the missing word

Rational expression — a fraction whose numerator and _____ are polynomials

Simplified form — the numerator and _____ have no common factors other than ±1

Excluded value — a value that makes a _____ 0

Part 1 · Notes and examples (fill in each note, then work the example)

Simplifying: Simplify: _____ first, then divide out common _____ (never terms).

Excluded values: Excluded values come from the _____ denominator.

Example 1

Simplify $\frac{x^2 - 2x - 15}{x^2 - 9}$. State the excluded values.

Multiplying and dividing: To multiply: factor, divide out, then _____. To divide: multiply by the _____.

Example 2

Find the product: $\frac{8x^3y}{2xy^2} \cdot \frac{7x^4y^3}{4y}$

Part 2 · Partner practice

Partner practice

Simplify $\frac{x^2 + 7x + 12}{x^2 - 16}$. State the excluded values.